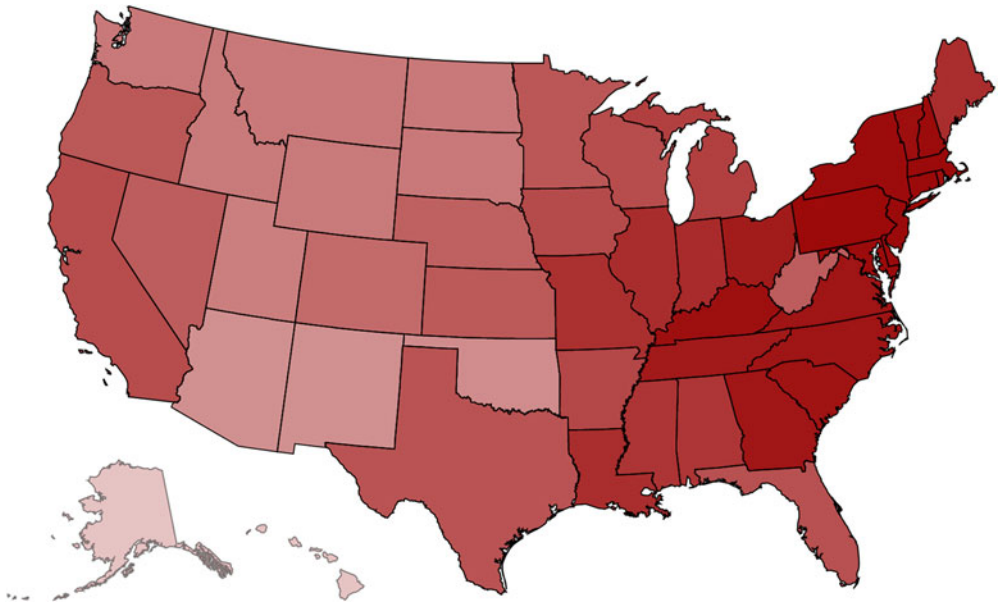


Figure 1.30 Geographic data stores the spatial geometry of objects, such as states. Each of the states in this image is represented as a separate feature with an array of values indicating its shape. Geographic data can also consist of points, such as for cities, or lines, such as for roads.

to a massive amount of publicly accessible geodata for any project. Geographic data has a few standards, but the focus in this book is on two: the GeoJSON and TopoJSON standards. Although geodata may come in many forms, readily available geographic information systems (GIS) tools like Quantum GIS allow developers to transform it into GIS format for ready delivery to the web. We'll look at geographic data closely in chapter 7.

1.4.5 Raw data

As you'll see in chapter 2, everything is data, including images or blocks of text. Although information visualization typically uses shapes encoded by color and size to represent data, sometimes the best way to represent it in D3 is with linear narrative text, an image, or a video. If you develop applications for an audience that needs to understand complex systems, but you consider the manipulation of text or images to be somehow separate from the representation of numerical or categorical data as shapes, then you arbitrarily reduce your capability to communicate. The layouts and formatting used when dealing with text and images, typically tied to older modes of web publication, are possible in D3, and we'll deal with that throughout this book, but especially in chapter 8.

1.4.6 Objects

You'll use two types of data points with D3: literals and objects. A literal, such as a string literal like `"Apple"` or `"beer"` or a number literal like `64273` or `5.44`, is straightforward. A JavaScript object, expressed using JavaScript Object Notation (JSON), isn't so